Circles Base Group Audit

Gnosis Ltd - Report by Côme du Crest

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Table of contents

- Table of contents
- Circles Base Group Audit
  - Scope
  - Context
  - Status
  - Optimisations and miscellaneous

Circles Base Group Audit

This document presents the findings of a smart contract audit conducted by Côme du Crest for Gnosis Ltd.

Scope

The scope includes all contracts within `src/base-group` of repo `aboutcircles/circles-groups` as of commit `0x1c32af9`.

Context

These contracts are a simpler rewrite of previously audited circles-cmg contracts.

Base groups are CirclesV2 groups that allows different humans to pool their personal circles (pCRC) into shared group circles (gCRC). The goal is to create a liquid currency backed by a number of different CRC.

The `BaseGroup` represent the address that will be registered as a group on the CirclesV2 Hub. The owner of the group can set the addresses (pCRC) trusted by the group. The `service` address can also added trusted addresses if they pass a membership check verified by calling `passesMembershipCondition(address)` on an array of membership conditions.

When trusting addresses, the `BaseGroup` will call the `BaseMintHandler` that mirrors the trust from the group to the added or removed address. The `BaseMintHandler` is thus capable of receiving pCRC tokens and triggers a mint for gCRC before sending the minted tokens to the sender of the pCRC. `BaseMintHandler` serves as a destination for circles pathfinding algorithm.

The `BaseGroup` registers on the hub using a `BaseMintPolicy` that simply returns true when called on `beforeMintPolicy()` and `beforeBurnPolicy()`.

The `BaseGroup` registers on the hub using a `BaseTreasury` contract as treasury. This contract will receive the pCRC tokens sent when minting gCRC tokens. The `BaseTreasury` contract trusts the group on deployment. When receiving tokens, the `BaseTreasury` will burn every held gCRC.

Contracts interactions are as follow: - Owner/service of BaseGroup sets trust from BaseGroup to members - Anyone can call `hub.groupMint(BaseGroup, collaterals, amounts, data)` to mint group tokens by sending trusted collateral. This calls `beforeMintPolicy()` on the registered `mintPolicies[BaseGroup]` from the hub which returns **`true`**. The only requirement is thus that the group trusts the input CRC tokens. This sends the collateral CRC to the treasury of the BaseGroup which is the `BaseTreasury` contract. This treasury upon receiving tokens will check on

the hub that the group trusts the tokens to accept them. - Alternatively, users can send their tokens to the `BaseMintHandler` that will trigger `onERC1155Received()` which calls `hub.groupMint()` for the user and transfer the received gCRC tokens back to the user. - Owners of gCRC can send them to the `BaseTreasury` along a transfer path to get pCRC tokens back. The `BaseTreasury` will burn the gCRC tokens purely for accounting.

Status

No issue have been found.

The report has been sent to the core developer.

Optimisations and miscellaneous

This part lists minor gas/code optimizations that shouldn't make the code less readable or improve overall readability. It also lists questions about unclear code segments.

Unused fee collector

The BaseGroup sets and stores a `feeCollection` that is never used. It can be removed.

https://github.com/aboutcircles/circles-groups/blob/1c32af9959b67e3ef3a5cb4d24adf968db7505a7/src/base-group/BaseGroup.sol#L154

BaseTreasury documentation typo

The BaseTreasury states "Treasury is registered as organization and trusts itself group". I think what is meant is "Treasury is registered as an organization and trusts its group …"

https://github.com/aboutcircles/circles-groups/blob/1c32af9959b67e3ef3a5cb4d24adf968db7505a7/src/base-group/BaseTreasury.sol#L12